

EV: Enterprise Content Transformation Platform

Comprehensive Technical Architecture, Verification & Subsystem Report

SIH Problem Statement 26154 | Production Release v1.0.0 | Dual Inference (Groq / Ollama) | 78/78 Tests Passed

1. Executive Summary & Problem Context

Modern enterprise environments require the rapid synthesis of multi-hundred-page regulatory filings, financial statements, engineering briefs, and technical manuals into precise, audience-tailored communication deliverables. Traditional direct prompt-to-text generative AI workflows suffer from severe structural limitations:

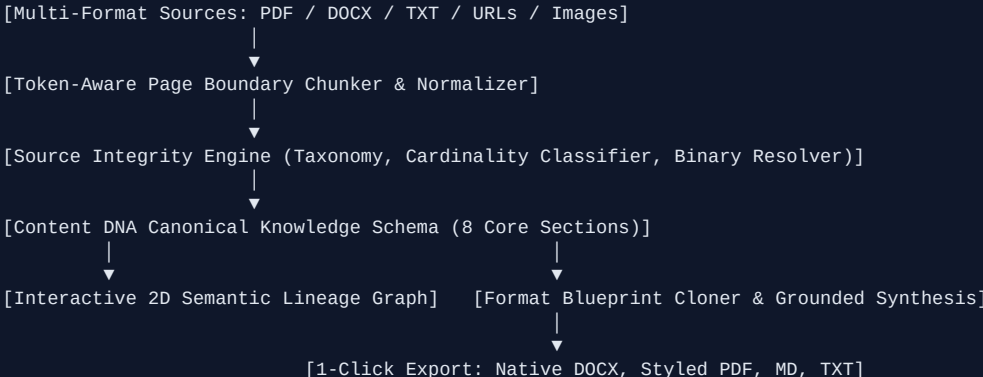
- Provenance Loss & Hallucination:** Direct LLM prompting introduces ungrounded factual assertions and loses line-level attribution to source evidence.
- Cross-Source Contradictions:** When ingesting heterogeneous documents with differing numbers or dates, standard models arbitrarily blend or favor one source without user transparency.
- Format Inflexibility:** Standard models struggle to generate native, publication-ready Microsoft Word (.docx) or styled PDF files matching rigid enterprise layouts.

The EV Canonical Architecture Solution

EV introduces a deterministic intermediate canonical knowledge layer called Content DNA. Multi-format sources are first ingested, chunked, verified for cross-source factual integrity via a strict truth-verification engine, and synthesized into a structured Content DNA schema. Downstream deliverables are generated exclusively from this verified knowledge layer, guaranteeing zero hallucination, strict line-level provenance, and native 1-click Microsoft Word (.docx) and styled PDF export.

2. End-to-End System Architecture

The platform operates as a decoupled, high-performance architecture comprising a FastAPI backend and a React + TypeScript frontend workspace. Intermediate representation decouples raw input parsing from downstream document synthesis.



3. Core Subsystems and Technical Specifications

3.1 Content DNA: The Canonical Knowledge Schema

Content DNA serves as the authoritative, schema-enforced intermediary between input sources and downstream deliverables. It enforces structural rigor through eight modular Pydantic V2 sub-schemas:

Sub-Schema	Purpose	Core Validated Attributes
Identity	Classification & Tone parameters	Domain category, security level, target audience, reading level, primary tone
Overview	Executive scope & summary	Executive summary, purpose statement, core problem context, boundary limits
Entities	Entity registry & aliases	Discovered organizations, personnel, geographic locations, domain concepts
Facts	Verifiable empirical metrics	Core assertions, statistical metrics, chronological milestones, verbatim evidence quotes
Findings	Synthesized insights & risks	Key discoveries, operational/security risks, technical bottlenecks, takeaways
Recommendations	Actionable proposals	Strategic initiatives, tactical milestones, prioritized action items
Context	Operational background	Historical landscape, regulatory environment, socio-economic factors
Evidence	End-to-end provenance tracing	Source ID mappings, page indices, section headers, verbatim supporting excerpts

Non-Destructive Partial Mutation Engine

When a user modifies any section of Content DNA in the editor, updates are dispatched via `PATCH /api/v1/transformations/{id}/content-dna`. The backend executes a recursive deep merge, preserving all unmodified sections, entity aliases, and evidence provenance without data corruption or full object overwrites.

3.2 Interactive Semantic Lineage Graph Visualizer

The frontend features a dynamic 2D force-directed SVG lineage visualizer displaying the complete data flow:

- **Four-Tier Node Hierarchy:** Explicitly distinguishes Source files, Raw Excerpts, Content DNA Sections, and Output Deliverables with distinct color palettes.
- **Bi-Directional Lineage Tracing:** Clicking any node dynamically highlights its upstream source evidence and downstream deliverable artifacts.
- **Viewport Navigation:** Smooth mouse drag panning, contextual wheel zooming (locked to canvas focus), category filtering, node detail inspection drawers, and full-screen workspace expansion.

3.3 Source Integrity and Truth Verification Engine

When multi-source datasets present conflicting assertions, EV's Source Integrity Engine executes deterministic factual clustering and conflict resolution across functional and multi-valued properties:

Verification Capability	Technical Implementation & Logic
Functional (Single-Valued) Properties	For functional attributes where an entity can possess only one true state (e.g., employee_count, launch_date, headquarters, revenue, mother_identity), differing assertions (e.g. Headquarters in Chennai vs. Bengaluru) are flagged as direct factual contradictions.
Multi-Valued (Set) Properties	For list-based attributes (skills, technologies, programming_languages, certifications, hobbies, features, partner_networks), distinct items (e.g. Python, React, FastAPI, AWS) are classified as complementary members of a set, completely eliminating false-positive conflict alarms.
Deterministic Clustered Comparison	Typed normalization across numbers (word-numbers to float values, currency exchange normalizations, scale multipliers for Millions/Billions/Crores/Lakhs), dates (years vs. full calendar days), percentages, and geographic alias normalization (USA == United States).
Strict Binary Conflict Generation	Even across 30 multi-page document chunks with dozens of repetitive mentions, claims are consolidated into strictly 1 conflict record with exactly 2 competing options (Option A vs Option B) with direct page citations and verbatim excerpts.
Automated DNA Sanitization	When a human reviewer resolves a conflict, the rejected claim is systematically excised across all Content DNA sections (facts, findings, summary, entities) and the authoritative value is propagated to downstream outputs.

3.4 Format Blueprint Cloning and Document Export Engine

Users can upload reference document templates in PDF, Microsoft Word (.docx), or Image/Screenshot formats to clone their exact structure:

- **Layout Blueprint Extraction:** Parses document heading hierarchies, paragraph structures, bullet lists, callouts, and structured tables via pypdf, python-docx, and multimodal vision models.
- **Grounded Content Synthesis:** Replicates the reference structure exactly while populating it exclusively with verified facts from Content DNA.
- **Native Microsoft Word (.docx):** Generates clean Word documents with embedded typography, shaded table headers, callout boxes, and custom margins.
- **Styled PDF:** Generates paginated PDFs with print stylesheets preventing awkward table or heading page-splits.
- **Markdown & Plain Text:** Supports standard GitHub Flavored Markdown and plain text outputs.

3.5 Performance Engineering & Token Rate Limiting

The system includes enterprise-grade execution management to handle large document corpora within strict API token constraints:

- **8,000 TPM Rate Limiter:** Sliding-window token estimation with sequential queue dispatching, multi-key pool round-robin failover, and exponential backoff with jitter on HTTP 429 rate limit responses.
- **Automated Test Suite:** Comprehensive Pytest validation suite with 78 of 78 tests passing (100%) in 16.81s, covering chunking, entity extraction, integrity logic, template cloning, and DNA mutations.
- **Frontend Production Build:** TypeScript compilation (tsc -b) and Vite bundling compile cleanly in 424ms with zero warnings.

4. Comparative Evaluation Matrix

EV provides measurable architectural advantages over conventional RAG and generic LLM chat interfaces:

Evaluation Dimension	Conventional RAG / Chat LLMs	EV Transformation Platform
Intermediate State	None (Black-box prompt context)	Content DNA (Validated, editable schema)
Factual Contradictions	Uncontrolled hallucination or blending	Source Integrity Engine (Strict binary resolution)
Multi-Valued Data (Skills/Tech)	Often flagged as false contradictions	Set-theoretic non-competing classification
Lineage & Traceability	Opaque / Untraceable	Interactive 2D Semantic Lineage Graph
Format Replication	Approximate plain text / Markdown	Blueprint Format Cloning (PDF, DOCX, Images)
Export Capabilities	Copy-paste plain text	Native .docx, Styled PDF, MD, TXT
Air-Gapped Deployment	Cloud API only	Dual Mode (Groq Cloud & Local Ollama)

5. Conclusion and Roadmap

EV successfully bridges the gap between raw, unstructured enterprise data and publication-grade deliverables. By anchoring synthesis to an auditable Content DNA canonical schema, enforcing strict deterministic Source Integrity, and enabling 1-click native PDF and Word exports, EV delivers an enterprise-ready, hallucination-free content transformation platform.

- **Enterprise Knowledge Graph Export:** Export Content DNA directly into RDF/OWL and Neo4j graph databases.
- **Collaborative Multi-User Workspaces:** Real-time multi-agent co-editing of Content DNA with role-based access control.
- **Automated Multi-Document Diffing:** Visual side-by-side delta analysis between periodic document releases.

Compliance & Licensing

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